

Universal properties from quantum many-body dynamics

Master's thesis defense

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Outline

- 1 Motivation
- 2 The model
- 3 Method: transverse contraction
- 4 Results
- 5 Conclusions

The ANNNI-type chain

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p \left(\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x \right) \right]$$

Equilibrium checks

From the echo to a transfer matrix

The entanglement barrier, and how we sidestep it

The corrected transfer-matrix column

Validation at the integrable point

The central charge: spectral route

The central charge: entropy route

The boundary operator spectrum

The four boundary pairs

Thank you

